Bless This Mess

I wrote a song called Bless This Mess, which i have chosen to record and collaborate with other students (Liz Castle). I'd like a female vocal for the second half of the song, leading into a duet for the final chorus.

Initial steps were to record the song vocal and guitar parts. I did this as a basic demo on my phone, which allowed me to share the lyric sheet and audio with Lizzie, and start the collaboration.

Song information

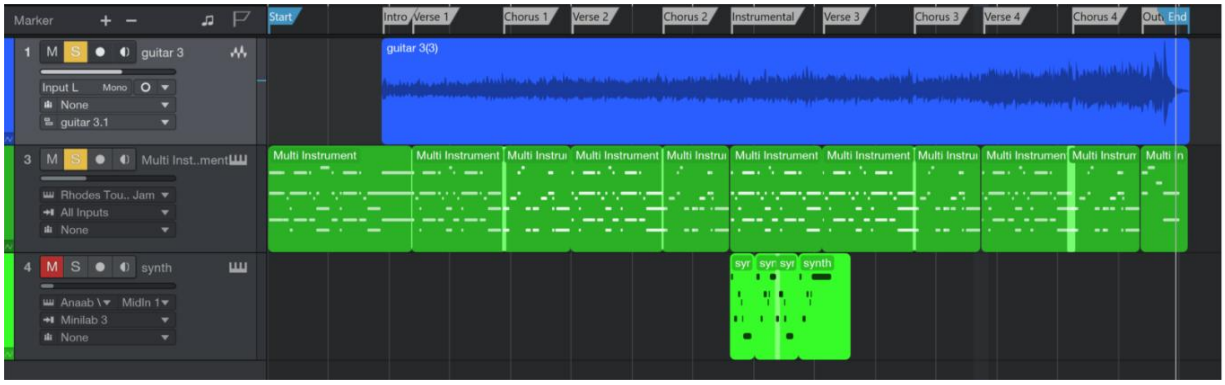
Time – ¾ @ 158bpm

Key – G major

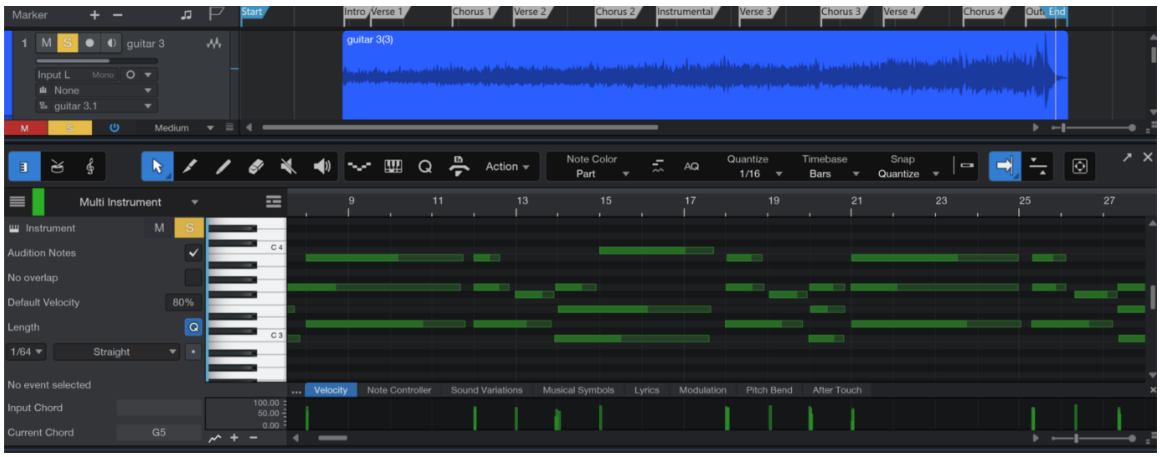
Recording Process

- I recorded the acoustic guitar parts via DI straight into my Quantum ES2 interface. These were captured using Studio One's *takes to layers* mode, which allowed me to start and stop freely, record multiple passes, and then comp together the strongest performances for each section.

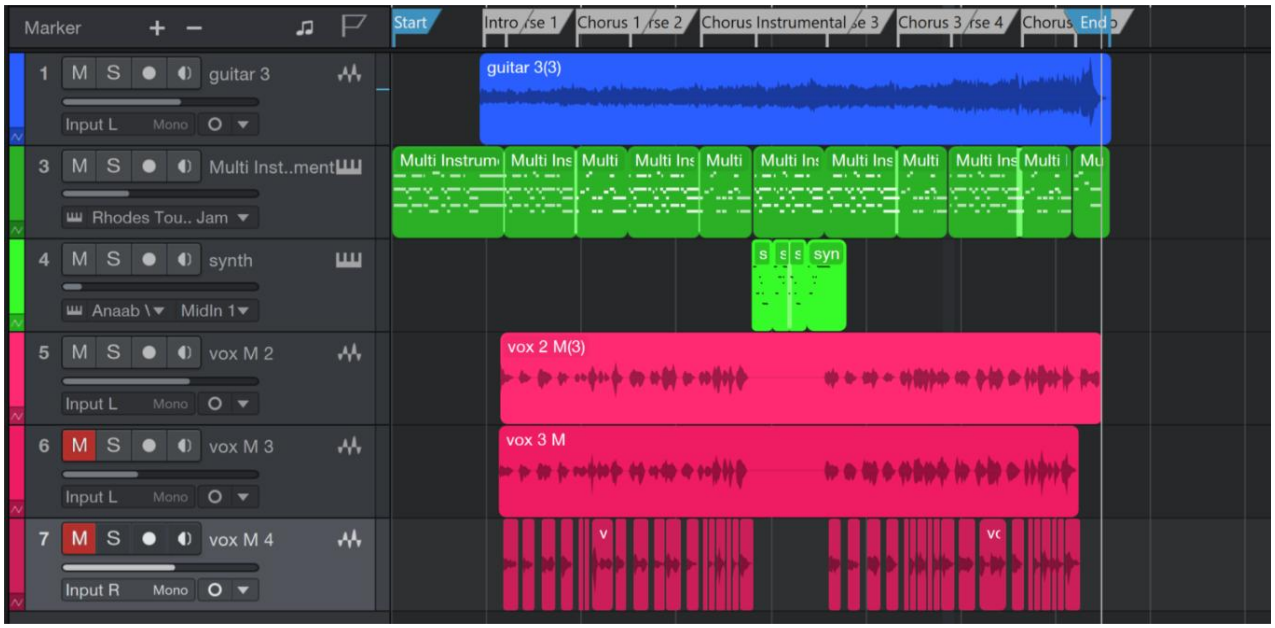
- I recorded the MIDI parts using my Arturia MIDI controller, beginning with the chord tracks. Because the song structure repeats (verse–chorus four times), I recorded one full cycle and then copied and pasted the sections to build the complete arrangement.

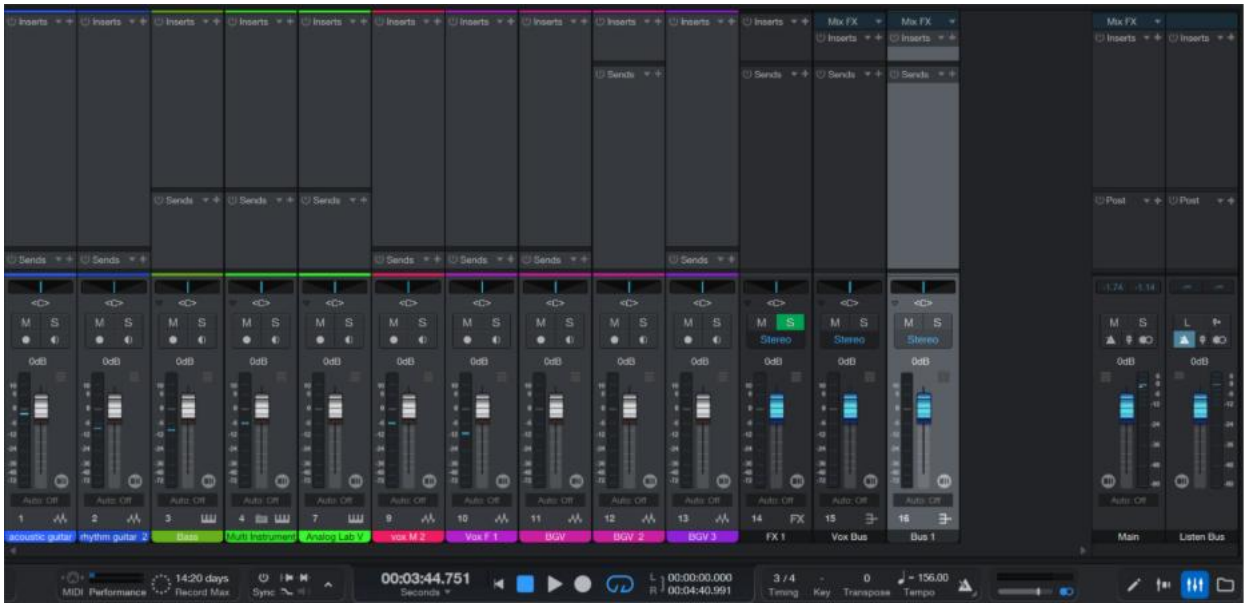


- I experimented with a range of Arturia instrument presets and selected sounds that provided the ambience and width I wanted for the track.
- For the instrumental section, I used the same MIDI workflow and chose a violin sound that suited the arrangement.
- I created the bass part using a virtual bass guitar instrument. I initially tried recording bass on my 6-string guitar tuned down an octave, but the physical noise from the strings made the performance unsuitable, so I opted for the cleaner MIDI bass instead.



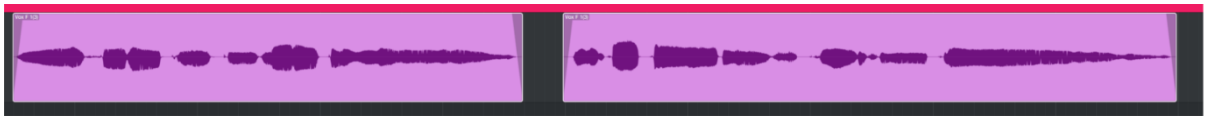
- Note velocity, length and timing can be altered in the edit section. This allows any mistakes to be corrected easily without having to play any parts of the track again
- I sent a demo and lyric sheet to Lizzie, who recorded an accompanying vocal and additional background vocals. Her harmonies and BGVs worked well with the arrangement, so I included them in the final track.
- For my own vocal, I sang into a dynamic microphone that was plugged into my acoustic amplifier. A condenser microphone was positioned against the amp's speaker tube to capture the amplified vocal tone. The condenser mic was connected to my Quantum ES2 interface to supply phantom power, and this signal was recorded directly into the DAW.



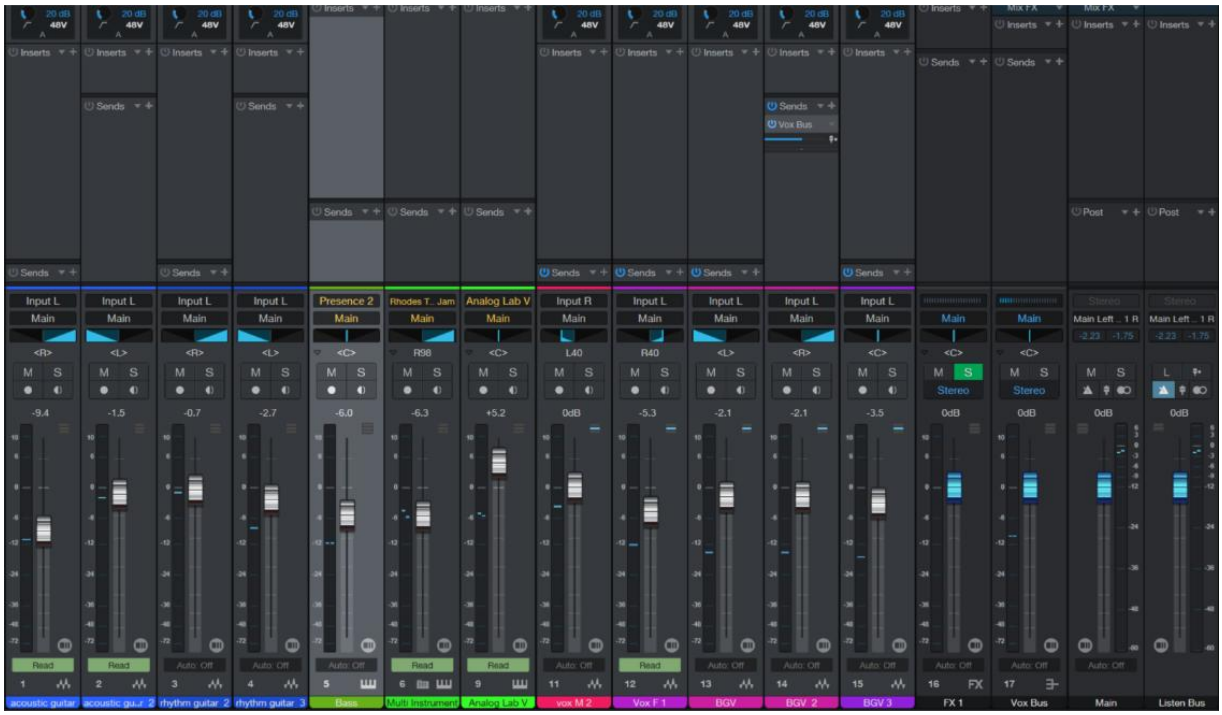


Mixing Process

- I applied short fade-ins and fade-outs (approximately 100 ms) at the beginning and end of each audio event. This helped eliminate the small transient clicks that can occur when an audio file starts or stops abruptly, ensuring smooth transitions and preventing unwanted artefacts in the mix.

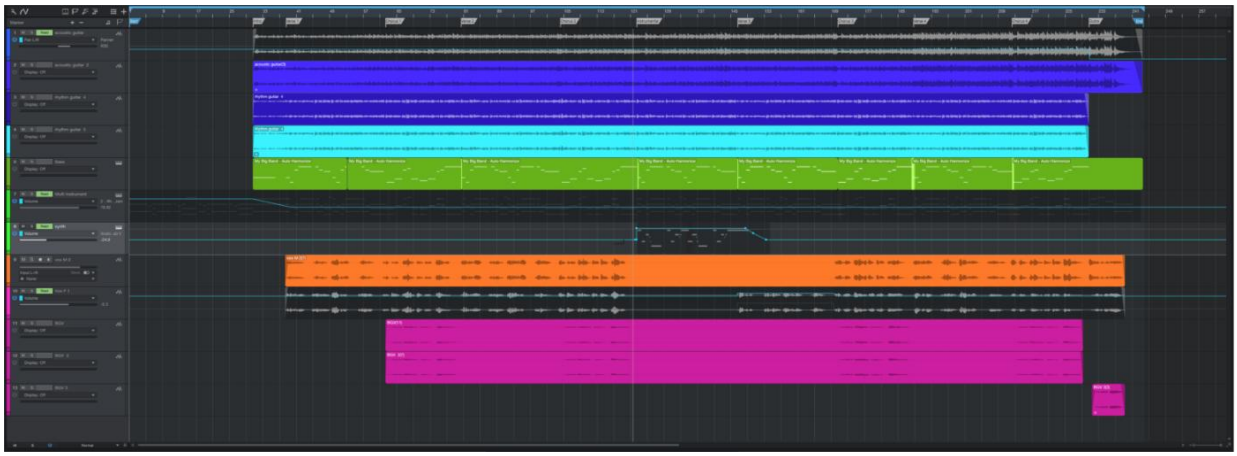


- I began the mix by establishing the overall balance of the track, setting the relative levels of the acoustic guitar, MIDI instruments, bass, and vocals to create a clear starting point.
- I used panning to create width and separation within the stereo field. The acoustic guitar and MIDI parts were positioned to complement each other, helping to open up the mix and prevent frequency masking.

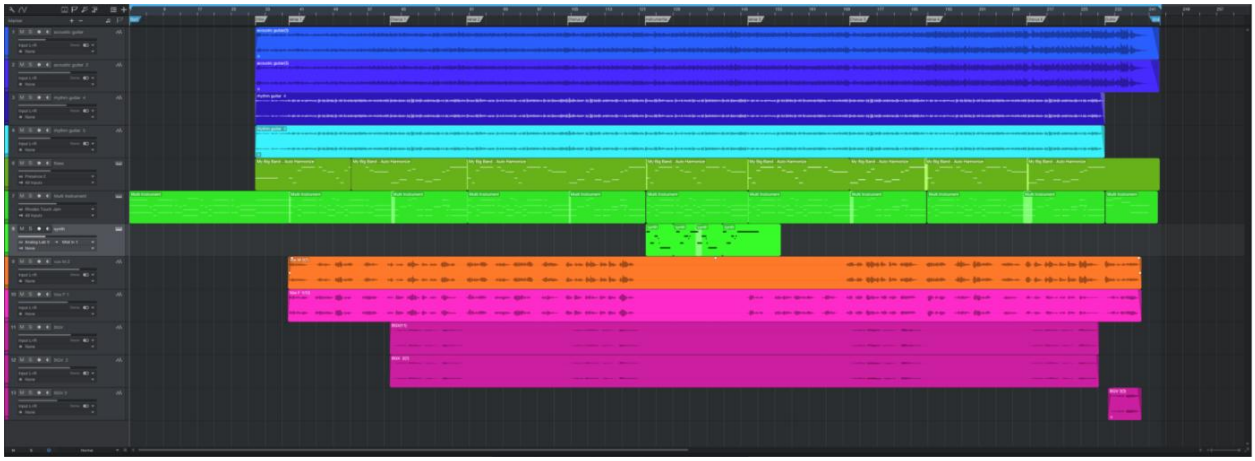


- I created a dedicated **vocal bus** and routed all vocal tracks to it. This allowed me to apply shared processing such as EQ, compression, delay, and reverb to both vocal parts, ensuring tonal consistency and a cohesive vocal sound.
- I also created an **FX bus** for the instrumental section, allowing me to apply reverb and other spatial effects to the MIDI violin and ambient elements without affecting the rest of the mix.
- EQ and compression were applied to the audio tracks where needed. The acoustic guitar received EQ to remove low-end muddiness and enhance clarity, while compression helped control performance dynamics. The bass part was EQ'd to clean up the low frequencies and ensure it sat well with the kick and guitar. The MIDI instruments required minimal processing, as their sound design already fit the arrangement.

- Once the tonal shaping was complete, I used reverb and delay to add depth and ambience. The instrumental parts were given subtle reverb to blend them into the background, while the vocals received a combination of reverb and a short delay to help them sit naturally within the mix.
- Automation was used throughout the session to maintain clarity and musicality. This included adjusting vocal levels between sections, controlling the intensity of the instrumental parts, and ensuring that key transitions remained smooth and expressive.



- Individual tracks were recorded as mono sources and then positioned within the stereo field using panning and spatial processing, resulting in a cohesive stereo mix



Vocal Noise Issue and Re-Recording Decision

After exporting an early mix and listening back on headphones, I noticed a persistent hissing noise in my own vocal recording that wasn't as obvious during tracking. The hiss became more pronounced once compression was applied, because the compressor was lifting the noise floor between phrases. This issue ultimately came from the way the vocal had been recorded through an acoustic amp, which introduced amp noise and reverb tail hiss into the signal. I tried several corrective approaches, including using a noise gate, adjusting EQ to reduce high-frequency noise, and rebalancing the vocal processing chain, but the hiss remained audible in quieter sections. To achieve a clean and cohesive final mix, I decided to re-record my vocal using the same DI method as the other vocal parts. This new recording blended far more naturally with Lizzie's vocals and resulted in a much cleaner, more professional overall sound.

Track by track processing

Track 1 – Acoustic guitar ghost track – panned slightly right and volume low to sit behind the rhythm guitar track but adds width to the mix, panned fully right for the outro as it the song is reduced to only the acoustic guitar and vocal, it was duplicated from track 2, given a 30ms delay and detuned –5 cents (100 cents = 1 semitone) to give a more natural feel

Track 2 – Main acoustic guitar – panned full left with EQ to make it brighter and compression added

Track 3 – Rhythm guitar 1 – panned full right to add stereo width, added slight delay using a pedalboard plug-in

Track 4 - Rhythm guitar 2 – panned slight left and turned volume low as ghost track behind the main acoustic guitar, adding stereo width, the track was duplicated from, nudged with a 30ms delay and tuned –5 cents (100 cents = 1 semitone) to give a more natural feel between the duplicated tracks

Track 5 – Is a lead guitar track i have disabled for this assignment due to time constraints, it is not completed yet and I'm not totally convinced it is required

Track 6 - Bass – instrument track played using a MIDI keyboard to sound like a bass guitar, EQ and compression added to remove muddiness, a noise gate was also added to limit clipping

Track 7 – Another MIDI track played as keyboard chords which are played in the background of the whole song. Due to the fact it is MIDI instrument track loaded in from Arturia presets; there are no plugins or audio processing added to this. I experimented with this and felt it did not add anything noteworthy to the song.

Track 8 – Again, a MIDI track made to sound like a violin for a small instrumental phrase in the middle of the song. As with track 7, there are no plug-ins added, although it is routed to an FX bus to add chorus and delay to the final note to boost it and give it a delayed fade out as the vocal comes back in. This can be seen in the automation view and the console send path on the track

Track 9 – Male Vocal – It was recorded through an amp with some reverb set up, so some EQ and compression applied to make the vocal clearer sounding and give room for more volume that was required in the final mix to make the track stand out

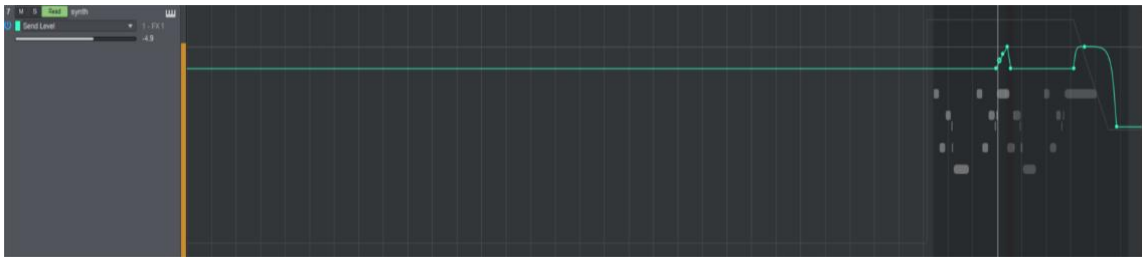
Tracks 10 – The main female vocal was automated to increase in volume for a short solo after the instrumental, i felt this give the song an amazing dynamic

Tracks 11 – 13 – The female track and all background vocal tracks were sent to the Vox Bus, created to add the same reverb, EQ, compression and some delay. All routed to the bus via a send, as seen in the console view above.

Vox Bus - The vocal bus plug in chain can be seen below. A gate is used first to reduce low-level noise between phrases, ensuring the vocal remains clean before further processing. This feeds into an EQ, which shapes the tonal balance by removing muddiness, controlling resonances, and adding clarity so the vocal sits naturally within the mix. A compressor follows to control dynamic range and keep the vocal consistently present, with a parallel blend adding fullness without sounding overly compressed. After the core tone and dynamics are established, a subtle analogue

delay provides depth and spatial support, helping the vocal sit more comfortably in the stereo field. Finally, a room reverb adds a natural sense of space and cohesion, blending the vocal into the overall arrangement without overwhelming the clarity of the performance.

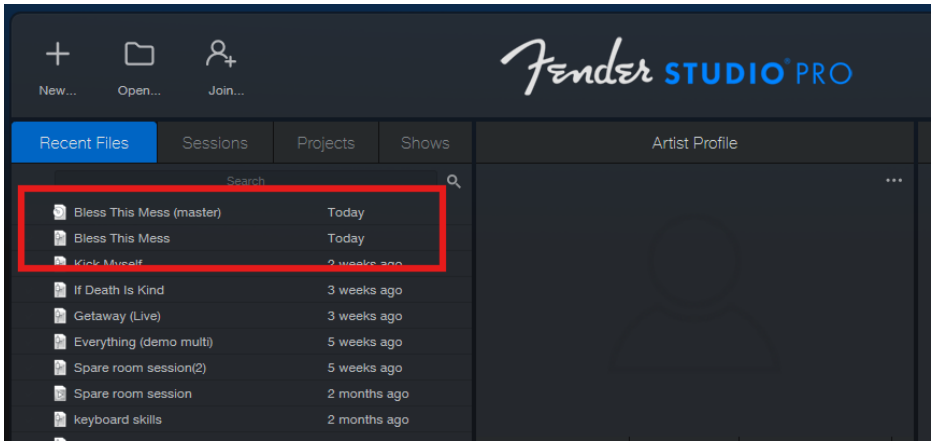
FX bus - I created a dedicated FX bus specifically for the instrumental solo section, allowing me to apply time-based effects without affecting the rest of the mix. Only the solo instrument was routed to this bus, where I used an analogue delay to enhance the longer sustained notes. This effect was automated so that it only appeared at key moments, adding a greater sense of expression and emotional impact to the performance. The delay helped the solo feel more spacious and dramatic, giving it a subtle lift that supported the musical phrasing without overwhelming the arrangement.

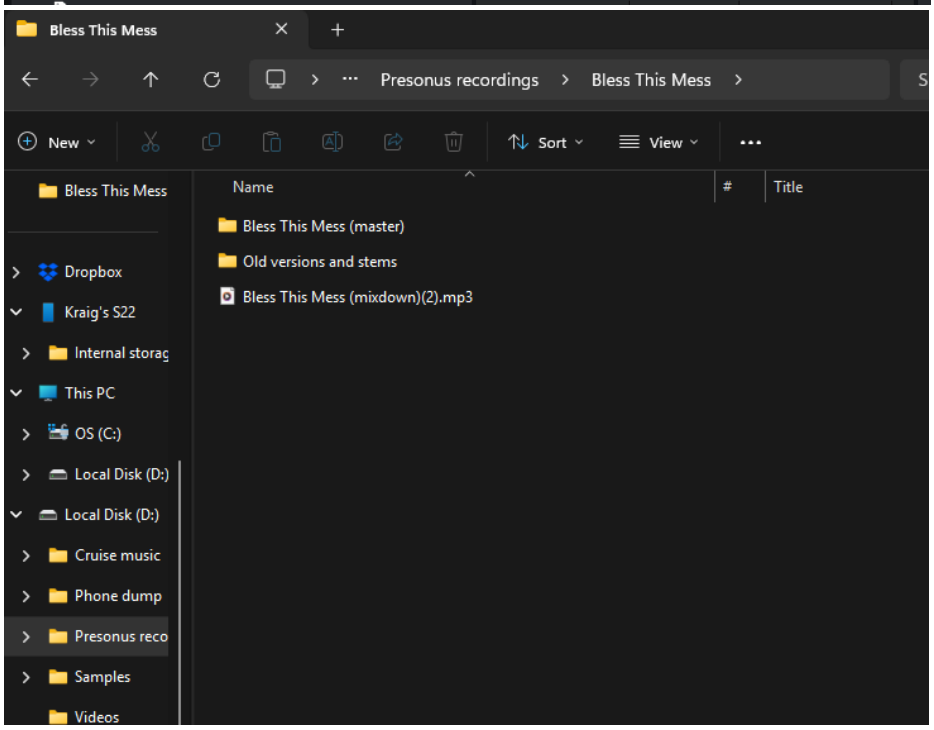


Mastering Process

- First i need to ensure my song is marked with start and end correctly as this is what will be exported to the project page (Studio Pro mastering suite)

- The file is exported to the project page and saved under a name that will make it easy to identify. This is saved onto the DAW and in my laptop (although i use a removable hard drive and save my music onto that)

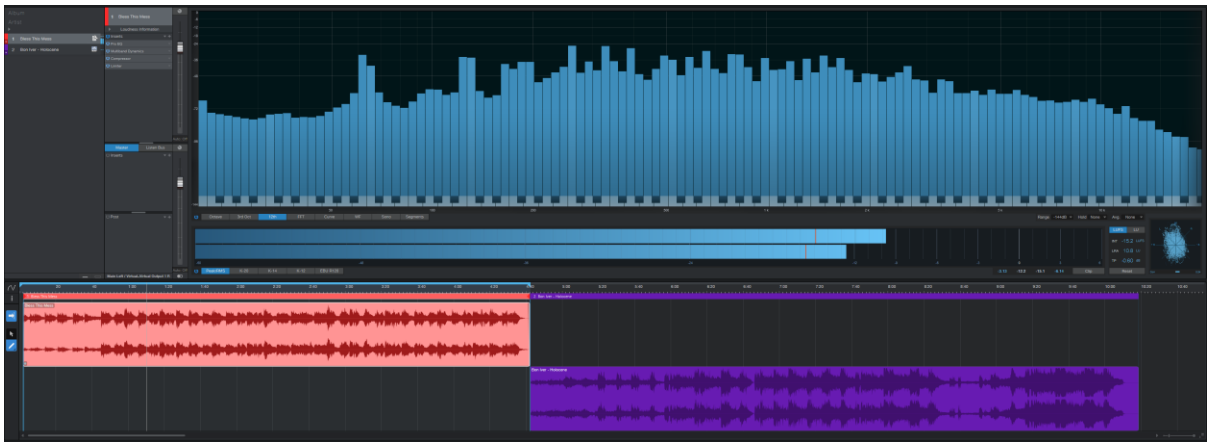




- The mastering page shows the song as a single waveform, i have imported a song called Holocene by Bon Iver to use as a reference track. It is a similar song

to mine, so it gives me a good indication if I'm doing things correctly as i enter the mastering stage.

- I've applied a fade in and out of 100ms to the waveform, for the same reasons it is done to each track during the mixing phase.
- I set up the mastering plug-in chain in a specific order, so each process is done in the correct order – i only add these plug ins to my own song, not the reference track. It is also prudent to turn the volume down (i dragged the gain down to around –10dB) on the reference track, as it will sound louder because it has already been mastered, this can cause confusion.
- EQ - Used to correct tonal balance by reducing muddiness in the low-mids and adding clarity in the high frequencies.
- Multiband dynamics - Used to control specific frequency ranges (particularly the low end) without affecting the entire mix.
- Compression - Applied lightly to gently glue the mix together and smooth overall dynamics.
- Limiter - Used to increase overall loudness to a final level while preventing clipping and distortion.

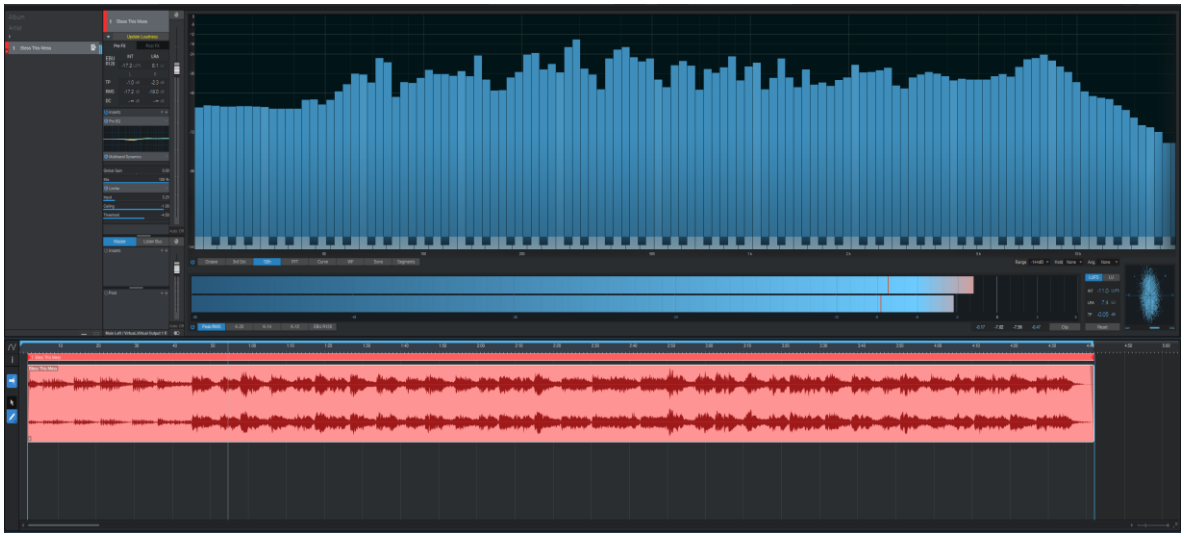


- The screenshot is taken while the track is playing to show all the information that is visible. The tracks are named (ref track on lower right), the plug-in chain can be seen in the right hand column. Top column is for that specific track, the bottom inserts column would be used to apply edits to a full album.
- Fine adjustment was made to the EQ to remove muddiness (LF reduced at 278Hz) and give the guitars and vocals more clarity (HF lifted at 9.2 kHz)
- Subtle lift of MF at 2kHZ to give a presence boost
- I used the LMF at 515HZ, i liked the clarity it brought to the guitar and vocal, it felt like it enhanced them and gave them more width

- The multiband dynamic compressor plug-in was then used to remove some harshness from the mid to higher frequencies and control the bass level

- Low band threshold will prevent the bass from moving above 12.72dB

- I found that lowering the gain on the HM and increasing it on the H really enhanced the vocals and helped them blend into the song more, making them feel more natural, this was a happy accident I discovered while playing around with different settings.

- Ultimately, the changes required on the multiband were minute, but they really blended the song together and I felt gave it a more 3D feel.
- I had a compressor in the plug-in chain, but experimenting with it never really added or removed anything from the song, so i decided to remove it altogether. The song has a naturally smooth dynamic and there are no huge transients, so further compression is not required.
- The final task was to apply a limiter to set the track loudness. It's important to listen while applying this as it is easy to choose higher volume over maintaining the feel and clarity of the song.
- The first step is to set the ceiling to –1dB, which will prevent the track from clipping
- Next is to listen to the song while gradually increasing the gain. I moved the knob manually to start with, then changed it in 0.25dB intervals. I found 3.25dB of gain to be the sweet spot, making the track come to life without distorting or removing clarity.

- Listening to the reference track on nearly every tweak and adjustment I made to my own song helped me confirm I was making improvements to the song without over-processing, I'm very happy with the way Bless This Mess turned out in the end.